

# Mach3 Software Walk-through

*This sequence begins after the equipment is correctly connected to all power/air required for operation, and are certain the communication between the PC and controller is correct.*

1. Open the Mach software (icon on desktop). Select the user profile (usually this is the name of the company or the owner of the machine) and press ok. The Mach screen should load.

*Please Note: There should never be another program open when Mach3 is running. If you do open another program, the Mach software should be closed and reopened.*

2. Turn the power to the CNC controller ON; make sure “E-Stop” is pulled out.

3. The user will notice your “RESET” button flashing at the bottom left hand side of your Mach3 screen. This means there is no communication between the machine and the PC. Click this button once and the LED should then stay a solid green, assuring there is communication.

*If the button continues to flash and the message line reads “External E-stop Requested” the user should check the cable connections and the power to the machine.*

4. The first thing the user should do after first powering on the equipment or if the software has been shut down is reference the machine by clicking on the “Ref All Axis” or “Ref All Home”, depending on the version of software, button located next the axis DROs ( Digital Read Outs). This process slowly looks for the physical limit switches (or proximity switches depending on machine model) located on each axis in order know its current position.

*When pressed, if the machine goes into a flashing reset with the message of “Limit Switch Triggered” you must hit the red “Stop” button and reset the machine again. Then make sure that the machine is not up against a limit switch (the machine may have to be jogged off of the limit switch) before you press the reference button again.*

5. Once the machine has referenced it will automatically move to the “Home” location and the soft limit light should then activate by turning yellow (top right of screen).

*Soft limits prevent the machine from running into the rails or stops on the X and Y axis. This does NOT prevent you from lowering your torch and then the jogging which could lead to a broken torch head. So be sure to always jog the cutter head high enough to clear any obstacles that may lead to a crash. The safest way to do this is by using the “GoTo Home” and “GoTo Zeros” buttons located at the top left of the screen.*

6. Load the program that the user wishes to cut by left clicking on the “Load G-Code” button located on the left hand side of the screen. Go to the location the .tap file was previously saved, which was created in Sheet CAM and open it. The user will see the “G Code” in the top left window of the screen and should see the tool paths of the part being cut in the preview window on the top right of the screen.

*There are two different part preview modes: Part View and Table View. The Table view zooms out far enough to view the entire parameters of the machine. Part view zooms in enough for the part to fill the window.*

7. Jog to the bottom left corner of your material (looking as the though the “Home” corner is the bottom left of the machine) by using your arrow keys on your keyboard (when holding shift + the arrow keys the machine will rapid jog, and CTRL + arrow will incremental jog). When you have the cutter head over the material, or wherever you wish to “Zero” the machine out at simply click the “Zero X” and “Zero Y” buttons located next to the axis readout numbers on the top of your screen.

8. After Zeroing the X and Y axis. Jog the Z axis down (page down on the keyboard) until the tip of the cutter is about .5” away from the material then press the “Zero Z” button located next to the Z axis DRO.

*Please Note: You do not have to zero the Z axis to the top of the material. The machine will automatically do a touch off before every pierce.*

9. Once all axis are zeroed out. You need to ensure that the “THC ON” light is flashing. If this light is not on or flashing the Torch Height Controller will NOT work which means the Z axis will remain at one constant height during the cut.

10. Make sure that the plasma cutter is setup to cut the material thickness that you are cutting (Air Pressure, Amperage, and Consumable Size). Also, put the ground directly on the material, NOT on the grates or machine frame.

11. The last thing before pressing the “Start” command is to “Regen Toolpath” or “Refresh Toolpath”, depending on the version of software you have. This refreshes the screen, and updates the machine to recognize where your updated zero location is. It is recommended this be done before starting a new cut file.

12. On the Torch Height Controller box make sure that the “Fine Cut” option is always turned off. The Operating Manual should give you a ballpark figure for “tip volts” to cut at. If the torch starts cutting too close to the material simply turn the knob towards the “UP” direction and the torch should raise up, and vice versa.

13. Everything should be setup correctly you are ready to begin the cut process by pressing “Start” or “Cycle Start” depending on the software version you may have. If anything seems to be incorrect you can stop the machine simply by pressing the “Stop” button, or pause (torch will remain on) with the “Feedhold” button.

# FAQ Troubleshooting Guide

## Mach3 software will not reset-

The Mach software not “resetting” occurs when after clicking on the “RESET” button on the lower left side of the screen, the Red/Green LED continues to flash instead of staying illuminated as a solid green bar when the machine is correctly “reset”.

- First, it sounds obvious but happens more often than you would suspect. Make sure that the control panel (metal box attached to the frame of the machine) is powered on AND the emergency stop is pulled out.
- Lack of communication will also prevent the machine from “resetting”. Double check that the 25 pin cables that run from PORT 1 (the factory port on your PC, most of the time runs vertical) to the control panel are securely screwed in.
- Also, if you did not purchase a computer system from ez Router Inc. the Port Address for the 25 pin parallel port may not match what is set in the Mach 3 software. You can locate and check your port address by going to your Device Manager on the PC that is running the machine (right click on my computer and select properties, then click the hardware tab, then select Device Manager). From the Device Manager you will expand the PORTS (COM and LPT section). If your cable is connected to the default port on the computer system (which is correct) you will right click on LPT1 and select properties. Select the resources tab, and you will see an I/O Range followed by an 8 digit number (ex: 0378-037F). You need to copy down the first four digits (0378) and close out of the Device Manager. Next, open the Mach software go to the upper menus and drop down the Config section. Select Ports and Pins, this will open a dialog box. You will see Port #1 and a Port Address right below that. Simply type in the four digits that we retrieved from the Device Manager (0378) and press “Apply” then the “OK” button.

## In the middle of a cut the plasma cutter simply stops cutting-

There are several reasons that this could happen here are a few things to look for:

- Check the consumables, sometimes even on a brand new set of consumable a piece of slag can get caught up in the torch head. Most of the time simply removing the consumable tapping it on a hard surface a few times and reinstalling it will take care of the issue. If this doesn't work you may have to replace both the electrode and tip.
- Check for obvious errors on your plasma cutter. Most plasma cutters will give you a fault light if there is an issue for things like air pressure too low, low voltage, etc. Then simply look in the troubleshooting section of the plasma unit's operating manual for further assistance.
- Watch the “ARC OK” light on your Torch Height Controller. If this light does NOT come on the ground loop coming from the plasma unit (ground strap) is not completing the loop. This can happen when the torch is moving to slow and there is no material to complete the circuit, or if the torch cuts off of the material. This is a safety feature to keep from breaking the head of your torch.

## If the plasma unit will NOT fire-

- If the plasma unit fails to fire during the program, check and see what line of code that you are stopped on (located on the left hand side of the Mach screen) then press the “STOP” button. Jog the torch up on the Z axis to a safe position and then try manually firing the torch (Torch On/OFF button located in the center of the screen). If the torch does NOT fire you need to check your consumables. Sometimes you can simply remove the tip and reinstall it and this will work, if it doesn't you may need to install a new set of consumables.
- Check the front of the plasma cutting unit for a fault. Most systems have a trouble shooting guide in the operation manual, look up the error the unit is given and repair accordingly.

*(A drop in air pressure is a very common issue)*

## **While cutting, the torch is either dragging the material or raising too high and loosing the arc-**

- Make sure that the “THC ON/OFF” button is flashing yellow ensuring that it is on. If this is not turned on the automatic torch height controller will NOT work.
- The arc voltage may need to be adjusted on the THC. If the cutting head is too close to the material, the tip volts (LED number displayed on the front of the THC) may be too low. Adjust these up by slowly turning the adjustment knob in the clockwise direction. This will raise the tip volts, which will cause the torch to move further away from the material. Turn the knob counter clockwise if the torch is too far away from the material, which will eventually cause you to lose your ARC OK signal.

*(Dragging the tip on the material can cause damage to your consumables)*

## **When cutting, the torch is not cutting through the material-**

- The “Pierce Delay” may be too short not allowing the torch to blow through the material before the cutter head begins moving on with cut. (The thicker the material the longer the Pierce Delay will be)
- The plasma unit’s amperage may need to be turned up higher. If the unit is maxed out, try slowing down the feed rate. The user may be cutting out of the unit’s recommended cutting capacity.

*(The plasma unit’s operation manual should list the maximum cutting capacity)*

## **How do I enter different parameters (feed rates, kerf width, pierce delay, etc)?**

- Mach 3 is simply the Control software for your Ez-Plasma. It simply does what the “G-Code” tells it to do. The user will use the Sheet CAM software to adjust the cutting parameters. Everything from air pressure to the recommended feed rates should be listed in Operator’s Manual for the Plasma Unit.